

HOLLOWGRAPH CALENDAR C2

Exact-date Microsoft Graph calendarView access associated with a covert bidirectional channel.

PRODUCTION CANDIDATE**CONFIDENCE: HIGH**

HOLLOWGRAPH / EXACT DATE

Detect successful calendarView GET requests that contain the exact source-observed date.

LOCAL IMPLANT

ENTRA APPLICATION

GRAPH API

2050 CALENDAR EVENT

TASK ATTACHMENT

REPORT DATE**22 July 2026****VERSION****v2.0****PRIMARY TELEMETRY****Microsoft Sentinel ·
MicrosoftGraphActivityLogs****ATT&CK / SCOPE****T1102.002 · T1071.001**

Independent defensive research. Source facts, analytical assessment, assumptions and detection logic are intentionally separated. Every query requires environment-specific validation.

01

EXECUTIVE ASSESSMENT

Decision-grade summary for threat research, detection engineering and SOC operations.

ASSESSMENT

EXECUTIVE SUMMARY

HOLLOWGRAPH Calendar C2

HOLLOWGRAPH repurposes a compromised Microsoft 365 identity and Microsoft Graph as a two-way dead drop. The analyzed implant queried calendarView for 13 May 2050 to retrieve tasking.

WHY IT MATTERS

Operational detection value

The exact far-future date is source-supported and observable in RequestUri when Graph Activity Logs retain query parameters. Destination controls otherwise see legitimate Microsoft Graph traffic.

High 01

ANALYTICAL CONFIDENCE



Focused 02

TELEMETRY COVERAGE



Production Candidate 03

OPERATIONAL READINESS



SOURCE-CONFIRMED FACT

Documented behavior

The analyzed sample issued GET requests to calendarView with a start date on 2050-05-13 and read tasking from an attached file.

ANALYTICAL ASSESSMENT

Interpretation

Exact-date access may provide a narrow campaign candidate; broader far-future activity belongs in hunting until legitimate applications are baselined.

DETECTION ASSUMPTION

Required condition

The tenant preserves startDateTime in RequestUri and collects sufficient identity and application context.

Evidence boundary

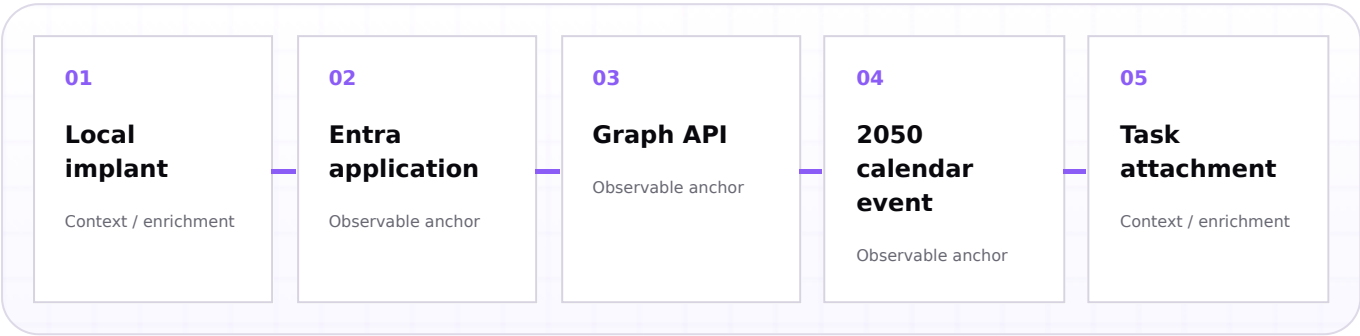
It does not expose event-body content, attachment names, encrypted data, implant presence or exfiltration success.

02

THREAT MECHANICS

Behavioral decomposition and observable evidence boundaries.

ATTACK FLOW



STAGE-BY-STAGE DETECTION VALUE

STAGE	BEHAVIOR	EVIDENCE EXPECTATION	USE
01	Local implant	Supporting process, file, identity or application context must be collected and correlated.	Context / enrichment
02	Entra application	Supporting process, file, identity or application context must be collected and correlated.	Context / enrichment
03	Graph API	The analyzed sample issued GET requests to calendarView with a start date on 2050-05-13 and read tasking from an attached file.	Primary anchor
04	2050 calendar event	Supporting process, file, identity or application context must be collected and correlated.	Context / enrichment
05	Task attachment	Supporting process, file, identity or application context must be collected and correlated.	Context / enrichment

Observable core

A match indicates successful Graph access to the exact far-future calendar range used by the analyzed sample.

Non-observable boundary

It does not expose event-body content, attachment names, encrypted data, implant presence or exfiltration success.

03

TELEMETRY & EVIDENCE MODEL

Minimum data contract required for a defensible implementation.

DATA CONTRACT

PRIMARY DATA SOURCE

Microsoft Sentinel · MicrosoftGraphActivityLogs

The primary platform is selected because it observes the strongest source-supported behavior with the most direct event relationship.

EVIDENCE BOUNDARY

A match is an investigative lead.

It does not expose event-body content, attachment names, encrypted data, implant presence or exfiltration success.

REQUIRED FIELD CONTRACT

TIER	DATA	PURPOSE	RELIABILITY
REQUIRED	TimeGenerated, RequestMethod, RequestUri, ResponseStatusCode, AppId, ServicePrincipalId, UserId, ClientAuthMethod, IPAddress, UserAgent	Core filtering, grouping and investigation output.	Documented schema; population must be verified.
OPTIONAL	Signer, hash, prevalence, file origin, software inventory, device group.	Benign separation and prioritization.	Environment-dependent.
ENRICHMENT	Threat intelligence, change windows, asset criticality and identity context.	Disposition and escalation.	Integration-dependent.

Completeness

Confirm table coverage, retention, field population and event latency before measuring prevalence.

Normalization

Validate case, path, quoting, URI encoding and command-line preservation against raw events.

Visibility gap

A different date or API workflow evades the rule; missing URI parameters makes the logic unusable.

DETECTION LOGIC

04

Source-grounded implementation with explicit assumptions and outcome boundaries.

PRODUCTION CANDIDATE

DETECTION OBJECTIVE

Detect successful calendarView GET requests that contain the exact source-observed date.

IMPLEMENTATION STATUS

Production Candidate - Untested implementation sketch

Syntax and schema are documented; local execution, field population and false-positive performance are not established.

KQL · DEMONSTRATIVE

```
// Title: HOLLOWGRAPH calendar C2 access via Microsoft Graph
// Status: Production candidate
// Confidence: High
// Purpose: Detect successful Microsoft Graph calendarView requests to the
//          exact far-future date observed in the HOLLOWGRAPH sample.
// Required table: MicrosoftGraphActivityLogs
// Required fields: TimeGenerated, RequestMethod, RequestUri,
//                  ResponseStatusCode, AppId, ServicePrincipalId, UserId,
//                  ClientAuthMethod, IPAddress, UserAgent
// MITRE ATT&CK: T1102.002, T1071.001
//
// Important:
// - This is a production candidate, not a production-validated rule.
// - RequestUri must retain the startDateTime query parameter.
// - The rule deliberately does not inspect event subjects, attachment names,
//   or request-body content because MicrosoftGraphActivityLogs does not
//   document those values as available fields.
// - Validate legitimate long-range calendar applications before deployment.
```

```
MicrosoftGraphActivityLogs
| where TimeGenerated >= ago(1h)
| where RequestMethod == "GET"
| where RequestUri contains "/calendarView"
| where RequestUri contains "2050-05-13"
| where ResponseStatusCode between (200 .. 299)
| project
    TimeGenerated,
    AppId,
    ServicePrincipalId,
    UserId,
    ClientAuthMethod,
    IPAddress,
    UserAgent,
    RequestMethod,
    RequestUri,
    ResponseStatusCode
```

What a match means

A match indicates successful Graph access to the exact far-future calendar range used by the analyzed sample.

What a match does not prove

It does not expose event-body content, attachment names, encrypted data, implant presence or exfiltration success.

05

TRIAGE & INVESTIGATION

Evidence-led workflow for defensible disposition.

SOC WORKFLOW

NUMBERED WORKFLOW

- 01

Confirm the match.
Review the raw event, timestamp, entity relationship and exact matching fields.
- 02

Establish legitimacy.
Identify the software or change owner, expected path, signer and authorized maintenance window.
- 03

Reconstruct origin.
Pivot to process, file, identity, network or application events before and after the match.
- 04

Measure prevalence.
Compare hashes, paths, commands, users and devices across the retained fleet.
- 05

Document confidence.
Separate confirmed evidence, inference, assumptions and unresolved gaps.

DECISION MODEL

Approved context → tune narrowly

Known owner, expected software path and repeatable behavior can support a documented exclusion.

Unknown context → deepen review

Rare or risky context without corroboration remains suspicious or insufficient evidence.

Corroborated activity → escalate

Malicious follow-on process, file, network, persistence or identity evidence supports incident escalation.

Host
Coverage, exposure, recent changes and neighboring alerts.

User
Session, intent, role and authorization context.

Process
Parent, child, path, command, signer and original filename.

File
Hash, signature, origin, prevalence and creation timeline.

Network
DNS, proxy, TLS and destinations following the event.

Identity
Sign-ins, token use, privilege and resource access.

OUTCOME	EVIDENCE PROFILE	ACTION
Benign	Approved and repeatable behavior with strong ownership.	Document and tune narrowly.
Suspicious	Risky context without decisive malicious corroboration.	Deepen endpoint or application review.
Likely malicious	Source-compatible match plus corroborated malicious activity.	Escalate incident response.
Insufficient evidence	Missing telemetry or unresolved ownership.	Retain evidence and validate gaps.

06

FALSE POSITIVES & VALIDATION

Operational controls required before alerting.

QUALITY GATE

FALSE-POSITIVE MATRIX

SCENARIO	LIKELIHOOD	TUNING ACTION
Calendar integration testing	High	Review owner, path, signer, prevalence and change context before exclusion.
Date-range boundary tests	High	Review owner, path, signer, prevalence and change context before exclusion.
Archival applications	Med	Review owner, path, signer, prevalence and change context before exclusion.
Synthetic QA activity	Med	Review owner, path, signer, prevalence and change context before exclusion.

Baseline plan

- Run retrospectively over a representative retention window.
- Measure volume by device, user, process, path and software owner.
- Inspect recurring legitimate patterns before adding exclusions.
- Preserve raw examples for peer review.

Validation procedure

- Confirm every required field against raw events.
- Execute only safe, authorized lab or historical validation.
- Compare query output with endpoint or application ground truth.
- Record success, rejection and sensor-failure criteria.

PROMOTION GATE

Hunting

Source-grounded hypothesis.

Candidate

Schema and baseline complete.

Production

Measured precision and runbook.

Optimized

Stable feedback loop.

Promotion criteria

Known alert volume, documented exclusions, representative validation, triage ownership, data-quality monitoring and measurable precision.

Failure condition

A different date or API workflow evades the rule; missing URI parameters makes the logic unusable.

07

ATT&CK, CONCLUSION & REFERENCES

Coverage statement, residual risk and document control.

CLOSURE

COVERAGE MAPPING

MAPPING	COVERAGE STATEMENT	BOUNDARY
T1102.002 · T1071.001	Partial behavioral or campaign-artifact coverage of the source-supported observable.	It does not expose event-body content, attachment names, encrypted data, implant presence or exfiltration success.

ANALYST CONCLUSION

Assessment, not certainty

The exact-date condition is assessed as a high-confidence production candidate, contingent on Graph URI retention and tenant-specific application review.

Residual risk

A different date or API workflow evades the rule; missing URI parameters makes the logic unusable.

REFERENCES

- 1. Group-IB - HOLLOWGRAPH: Turning Microsoft Graph into a Covert C2 Channel: <https://www.group-ib.com/blog/hollowgraph-microsoft-365/>
- 2. Microsoft Learn - Microsoft Graph activity logs overview: <https://learn.microsoft.com/en-us/graph/microsoft-graph-activity-logs-overview>
- 3. Microsoft Learn - MicrosoftGraphActivityLogs table reference: <https://learn.microsoft.com/en-us/azure/azure-monitor/reference/tables/microsoftgraphactivitylogs>
- 4. T1102.002 - Web Service: Bidirectional Communication: <https://attack.mitre.org/techniques/T1102/002/>
- 5. T1071.001 - Application Layer Protocol: Web Protocols: <https://attack.mitre.org/techniques/T1071/001/>

DOCUMENT CONTROL

VERSION	DATE	STATUS	OWNER
v2.0	22 July 2026	Production Candidate	Detection Engineering

Change summary

Migrated to the permanent Threat Research Editorial HTML master; technical query and evidence classification preserved.

Publication control

No customer data, credentials or environment-specific internal indicators are included. URLs are defanged in the publication artefact.